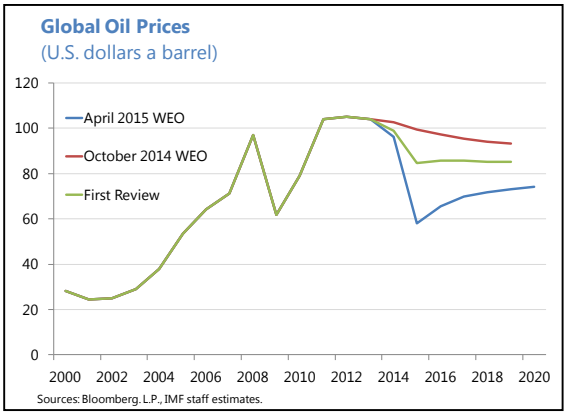
Annex I. The Macroeconomic Impact of the Decline in Global Oil Prices

Global oil prices have declined dramatically since mid-2014, reflecting additional supply on global markets combined with a weaker outlook for global demand. *This annex considers the potential impact of the decline in the world price of oil on Grenada. Grenada, as a net oil importer, is expected to benefit from the decline in oil prices through positive growth effects as well as an improvement in the external current account balance.*

A. Developments in Global Oil Prices

1. **Global oil prices have declined by more than 40 percent since mid-2014.** IMF analysis suggests that while demand weakness in some major economies, particularly emerging market economies, has played a role in the oil price decrease, that greater-than-expected supply has been the dominant factor.[1,2] With the demand and supply factors driving the decline in the world price of oil expected to persist, the IMF forecast for oil prices has been revised down significantly since the October World Economic Outlook (WEO) baseline and since the staff report for the first review of Grenada's ECF-supported program.[3] The April 2015 WEO baseline oil price forecast for 2015 is 42 percent below that of the October WEO baseline and 31 percent below that underlying the macroeconomic framework for the first review. The decline in prices is expected to persistent well into the medium term, with oil prices in 2020 expected to remain about 20 percent below the October 2014 WEO baseline.



B. Expected Real Effects

2. **Lower global oil prices are expected to provide a boost to economic activity in Grenada through three channels:**

- **Domestic demand**: As an oil importer, Grenada could benefit directly from the decline in global oil prices, as lower prices reduce the domestic cost of transportation and electricity and as the resulting increase in real disposable income stimulates consumption and economic activity.

[1] IMF April 2015 World Economic Outlook.

[2] The supply effect has included a steady rise in production in countries not belonging to the Organization of Petroleum Exporting Countries (OPEC), especially the United States; the faster-than-expected recovery of production in some stressed OPEC producers; and especially OPEC's November 2014 decision to maintain production levels despite the sharp decline in prices.

[3] The IMF oil price forecast is based on a simple average of the spot prices of Dated Brent, West Texas Intermediate, and the Dubai Fateh crude prices.

However, given the structure of Grenada's economy, the increased consumer spending is likely to be largely incident on imported consumer goods rather than domestic production.

- **External demand**: Grenada could benefit indirectly from the real income effect in other oil importers through increased demand for Grenada's tourism services. Tourism demand could also increase as the cost of air travel is reduced.
- **Competitiveness (supply):** The lower cost of production and associated increase in firms' profit could also stimulate investment. Lower global oil prices may reduce production costs for Grenada by more than its competitors with a more diversified energy base.

3. **A vector autoregression (VAR) model similar to that in Cashin and Sosa (2013) is used to estimate the impact on real GDP of an oil price shock.**[4] The VAR model with block exogeneity constraints includes five endogenous variables and one exogenous variable. The endogenous variables are divided into two blocks: (i) one domestic, including growth in real GDP and the real effective exchange rate; and (ii) one external, including growth in the world oil price and advanced economies, as well as the real interest rate in advanced economies. Finally, the exogenous variable captures the effect of natural disasters with a dummy variable constructed as in Acevedo (2014).[5] The model suggests that the elasticity of real GDP to a 1-standard deviation shock to the price of oil (equivalent to a 32 percent increase in the price of oil) is -0.3 on impact.[6] This suggests that the roughly 30 percent decline in the oil price forecast for 2015 since the first review could increase economic activity in Grenada by 0.3 percentage points in 2015.

4. **The economic outlook for 2015 has been revised upward, with the improvement partly attributed to the decline in global oil prices.** Growth is now expected at 1¾ percent in 2015, up from 1.1 percent at the time of the first review, with 0.3 percentage points of the upward revision attributed to the decline in the outlook for world oil prices. The remainder of the upward revision for 2015 is reflective of stronger prospects for the tourism sector, as well as stronger momentum in agriculture than previously anticipated. The outlook for 2016 has also been revised upward by 0.3 percentage points, to 2.0 percent. The medium-term forecast remains unchanged, reflecting an unchanged view on potential output.

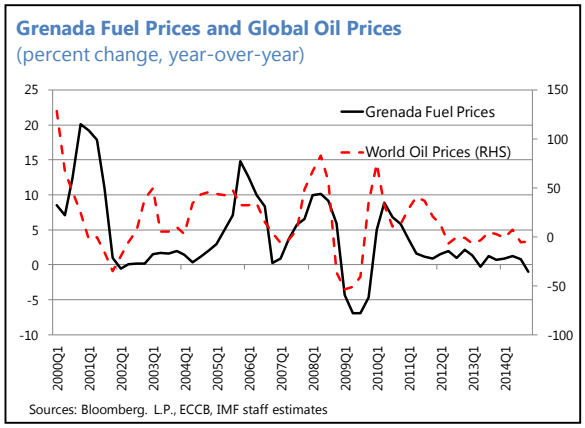
5. **The decline in world oil prices is expected to contribute to continued deflation in 2015, but the pass-through to core prices is expected to be limited.** The decline in the world price of oil is expected to be passed through to retail fuel prices in Grenada. Retail fuel prices are

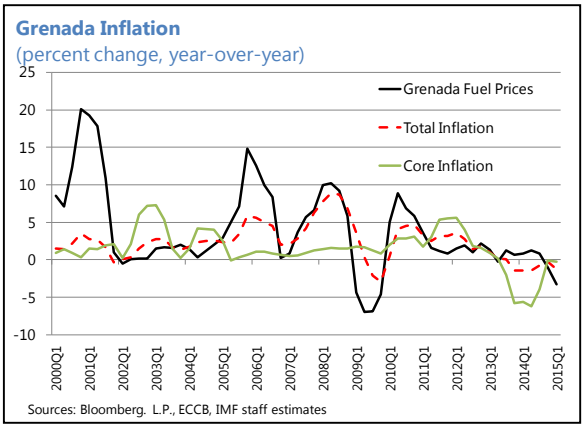
[4] Cashin, P. and Sosa, S., 2013, "Macroeconomic fluctuations in the Eastern Caribbean: The role of climatic and external shocks," *The Journal of International Trade and Economic Development*, Vol. 22, pp. 729-748.

[5] The dummy variable captures the effects of storms, floors, earthquakes, volcanic activity and droughts. Acevedo, S. (2014), "Debt, Growth and Natural Disasters: A Caribbean Trilogy," IMF Working Paper 14125 (Washington, DC: International Monetary Fund).

[6] At the same time, the model suggests that the lagged effects could be substantial with a one standard deviation decline in the price of oil expected to increase growth by 2.4 percentage point the first year following the shock. However, it is important to note that the confidence interval (-3.7 to -1.1 percent) is very large, suggesting significant variability in the response of growth to an oil price shock.

based on the f.o.b. prices of imported refined fuel products in the previous month, with complete pass-through of the change in the imported prices to the final consumer. Indeed, the correlation between growth in Grenadian fuel prices and lagged world oil price inflation is 0.6.[7] While developments in total inflation are closely related to developments in fuel prices (contemporaneous correlation of 0.6), the correlation between fuel prices and core inflation is negative (-0.1), suggesting that movements in fuel prices are not passed through to core inflation.





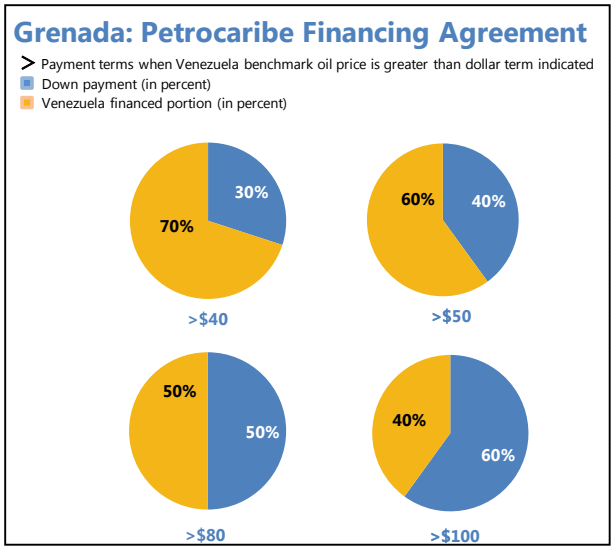
6. **The strong pass-through of world oil prices to total inflation and limited pass through to core inflation is confirmed by regression analysis.** In particular, pass-through to core inflation would be expected to be limited, as world oil prices are not a significant determinant of core inflation in Grenada. However, based on the estimated relationship between total inflation and the world oil price, a 30 percent decline in the world price of oil would be expected to reduce total inflation by 0.6 percentage points. This relationship has been factored into staff's re-evaluation of the inflation outlook. However, the impact of the decline the world price of oil on consumer prices in Grenada is expected to be offset by the increase in the Petrol Tax in March 2015 by EC$1. On balance, total inflation (average annual basis) is now expected to decline by 0.4 percent in 2015 compared to the moderate inflation (1.3 percent) expected at the first review.

C. Balance of Payments Effects

7. **The current account deficit is expected to improve with a lower fuel import bill**, although this effect would be somewhat offset by a modest pickup in import volumes due to the positive real income effect. On a net basis, Grenada's current account deficit is expected to improve by 2.7 percent of GDP in 2015 and by about 1½ percent of GDP over the medium term, on account of lower oil prices. For the overall balance of payments, this positive effect on the current account will be somewhat offset by a change in the financing terms of Grenada's Petrocaribe arrangement with Venezuela. With global oil prices expected to remain in the $50–$60 per barrel range over 2015, the share of Grenada's oil imports financed under its Petrocaribe arrangement is expected to

[7] Fuel CPI is not reported separately in Grenada. Staff defines fuel price inflation as transportation costs plus housing, utilities, gas and other fuels.

drop from 60 percent to 40 percent (see text figure). As a result, Petrocaribe financing is expected to decline by ½ percent of GDP to 1.2 percent of GDP in 2015, reflecting both the expected lower value of imports as well as the smaller share of Petrocaribe financing. On net basis, the impact on the external position is notably positive (2¼ percent of GDP improvement relative to the first review, due solely to the lower oil prices) and would remain marginally positive even under a complete disruption of Petrocaribe financing.



D. Fiscal Effects

8. **The decline in world oil prices is expected to have a net negative impact (about ½percent of GDP) on the fiscal balance due to reduced Petrocaribe grants for social spending, but this effect is already being offset through policy measures:**

- Petrocaribe grants, which covered about 40 percent of total social spending (or about 1 percent of GDP) in 2014, are projected to fall by half in 2015 as a result of the lower oil prices and the subsequent lower share of concessional financing available under the arrangement. The authorities had to reallocate resources elsewhere to maintain spending on the social programs supported by these grants.[8] To the degree that any further decline in oil prices may precipitate a disruption to Grenada's Petrocaribe arrangement, this remains an important risk to the fiscal position. An abrupt withdrawal of this funding could potentially derail these social projects unless the government could effectively identify additional grant resources, including receipts from the CBI program.
- While the government transportation and energy costs are expected to fall somewhat with the lower oil prices, additional policy measures have been taken by the authorities to neutralize the negative impact from the reduced Petrocaribe grants, including an increase in the fuel tax (by EC$1 to EC$4 per imperial gallon on gasoline and diesel) and lowered expenditure allocations relative to the budget.
- Lastly, lower oil prices could potentially reduce regional savings (generated in oil-producing countries), on which many tourism-dependent countries have relied on for fiscal financing during the global downturn, with a corresponding tightening of the financing conditions.

Annex II. Progress in the Restructuring of Grenada's Public Debt and Updated Debt Sustainability Analysis

[8] Particularly in the areas of education, employment support, infrastructure and housing.

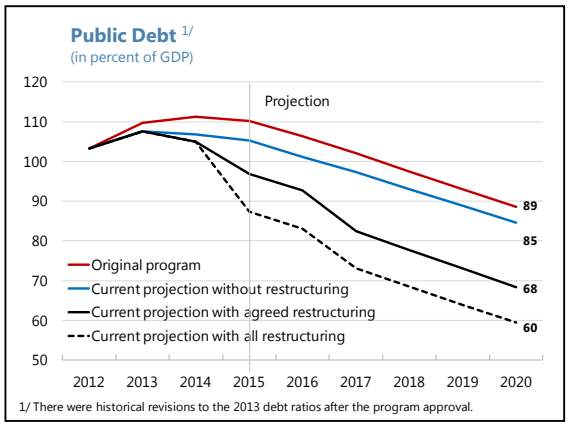
Significant progress has been made in the comprehensive restructuring of Grenada's public debt. The authorities reached agreement with creditors holding over half of Grenada's restructurable debt, which will reduce public debt by 13 percent of GDP by 2017 compared to the projected trajectory without the restructuring. This agreement marks a critical milestone toward debt sustainability and restoring orderly relations with financial markets. Moreover, significant progress has also been made in the clearing of Grenada's arrears. These developments support an improvement in Grenada's external risk of debt distress, but the classification of debt in distress is maintained until all debt restructuring elements have been completed and external arrears have been eliminated.

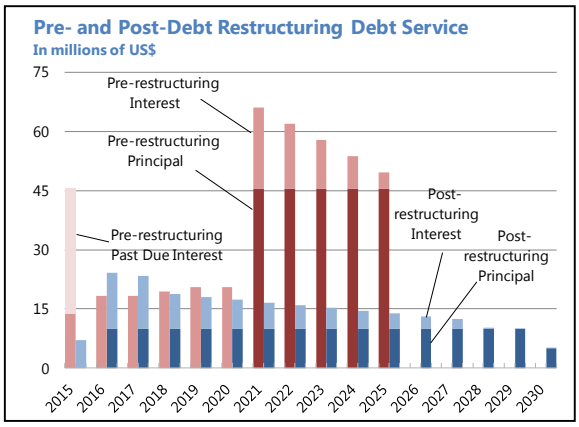
A. Progress in the Restructuring of Grenada's Public Debt

1. **Important progress has been made in the restructuring of Grenada's public debt across all creditor groups, with debt restructuring agreements reached with some private and bilateral creditors:**

- ***Private creditors:***
  - **On April 8, 2015, the Government of Grenada announced an agreement on key debt restructuring terms with the creditor committee that represents the majority of its private bondholders.** The in principle agreement covers the US$ and EC$ bonds issued in the 2005 restructuring (the "2025 bonds"), which account for close to one-third of total debt. As of end-March 2015, the total sum owed by Grenada to the 2025 bondholders (including past due interest) was US$296.8 million or 33 percent of GDP, with 75 percent denominated in US dollars.
  - **The agreed financial terms include a phased 50 percent nominal principal haircut.** Half of the principal reduction will take effect upfront and half upon completion of the sixth review of Grenada's ECF-supported program (expected in 2017). Past due interest will be capitalized, but will not receive a haircut. The restructured debt will have a maturity of 15 years, a fixed interest rate of 7 percent, and 29 equal semi-annual principal repayments beginning early 2016. The overall terms imply an NPV reduction of the affected debt of 54 percent, assuming a 12 percent exit yield. Grenada expects to launch an exchange offer for the 2025 bonds once the legal aspects of the new instrument, which will replace the existing bonds, have been finalized.
  - **The agreement also includes the sharing of potential receipts from the recently launched CBI program.** The share of CBI receipts going to the creditors will vary positively with the magnitude of the annual inflows—which is highly uncertain—with cumulative flows capped on an NPV basis. The sharing of the CBI receipts will start only after the ECF-supported program and will exclude the first US$15 million in annual CBI revenue. This feature will effectively allow the creditors to claw back some of the haircut in the event that the CBI program is highly successful. The authorities and the committee are also discussing the inclusion of a 'hurricane clause' in the new instruments, to allow a delay in debt servicing following hurricanes.

- **High bondholder representation on the creditor committee is expected to help satisfy the collective action clause voting threshold and ensure universal participation in the bond exchange.**[9] The creditor committee represents about 76 percent of the total outstanding 2025 bonds— one of the few restructuring cases with such high representation. Support of these creditors will help achieve the 75 percent voting threshold required by the collective action clauses in the 2025 bond contract to ensure 100 percent participation in the bond exchange, which is expected to be concluded around mid-2015.[10]
- **The agreement is a significant milestone towards restoring debt sustainability and orderly relations with the markets.** The restructuring of the 2025 bonds will reduce public debt by 13 percentage points (to 81 percent) of 2017 GDP by that year compared to the projected trajectory without the restructuring. The agreement will also provide notable cash flow relief, with the NPV saving on the 2025 bond cash flows of about 46 percent.





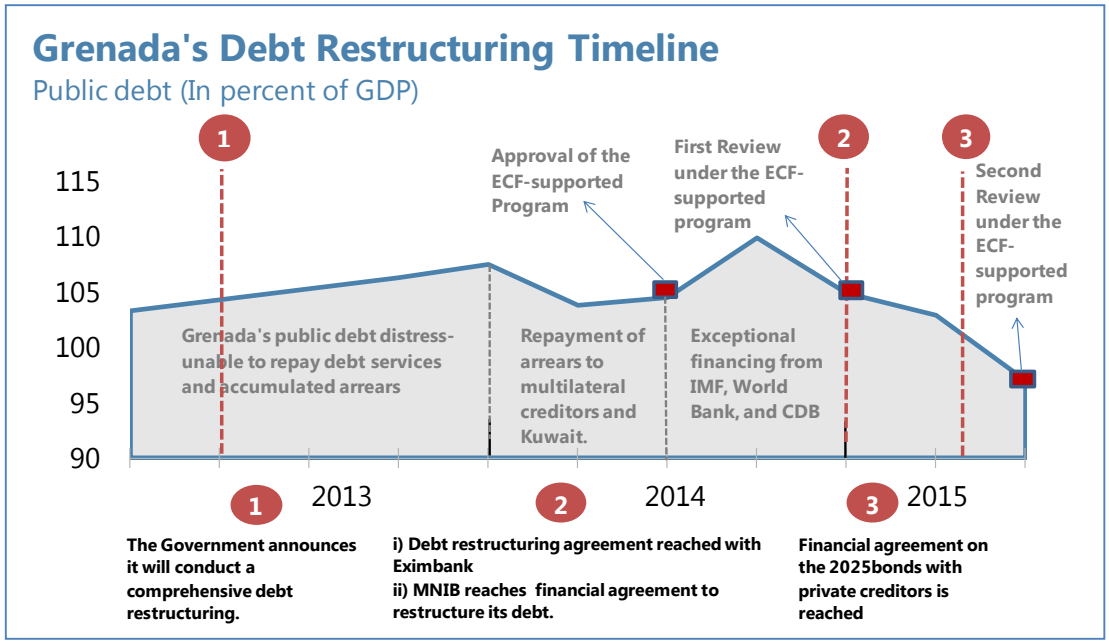
- ***Bilateral official creditors***: On January 7, 2015, the Government announced completion of a restructuring agreement with the Export-Import Bank of Taiwan Province of China (Eximbank). The agreement resulted in an immediate principal reduction of 47 percent on Grenada's liabilities of US$36.5 million (2 percent of GDP) to Eximbank.[11] A further 3 percent principal reduction will take effect upon completion of the 6th review under the ECF arrangement (expected in 2017). The restructured debt has a coupon rate of 7 percent, a grace period of 3½ years, and a final maturity of 15 years. The overall terms resulted in an NPV haircut of 62 percent and a decline in the 2014 public debt stock by about 2 percent of GDP. Importantly, in February 2015, Eximbank's legal proceedings in New York court to recover payments outstanding since the 2005 debt restructuring on the basis of the *pari passu* contractual clause included in the loan agreement were dismissed with prejudice (i.e. Eximbank cannot re-file the litigation).

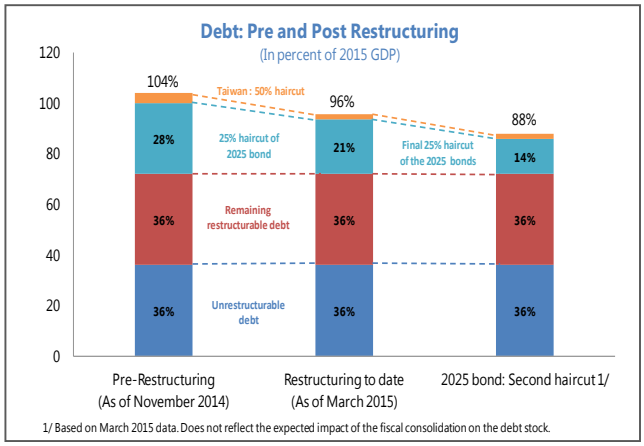
[9] The largest bondholders on the committee include Franklin Templeton, GMO, Greylock, and T Rowe Price.

[10] The government is in the process of agreeing to a comparable long-term flow rescheduling with the National Insurance Scheme, which holds about 4 percent of GDP of debt in the 2025 bonds.

[11] While the announcement was made in January 2015, the debt restructuring transaction took place in December, 2014.

- ***Central government guaranteed debt:*** As part of its reform efforts, Grenada's Marketing and National Importing Board (MNIB), a state-owned enterprise, has reached financial agreement with its private creditor to restructure US$5.4 million in government guaranteed debt. The agreement implies a reduction of the principal outstanding by 47 percent, and, importantly, includes the removal of the guarantee from the restructured debt. MNIB also made an upfront payment of US$1.5 million, resulting in a reduction in the stock of MNIB's government guaranteed debt to US$4 million (0.5 percent of GDP) at end-2014. Upon finalization of the restructuring, the stock of government guaranteed debt will decline by 8 percent (0.5 percent of GDP). Guaranteed debt of other parastatals is being included in the overall debt restructuring negotiations.
- ***Remaining Creditors:*** The authorities are continuing debt restructuring negotiations with their remaining creditors. The Paris Club is tentatively scheduled to discuss the treatment of its claims on Grenada in September 2015. The government has also contacted all non-Paris Club bilateral official creditors, and negotiations are ongoing with domestic creditors.





Financial Terms for the Restructuring of the US$ and EC$2025 Bonds

Financial Terms	April 2014 Published Terms		Export-Import Bank of Taiwan Province of China	2025 Agreed Terms
	Option 1	Option 2		US$ and EC$2025 bond 1/
Coupon	6.5% p.a.	5% p.a.	7% p.a.	7% p.a.
Tenor	15 years	20 years	15 years	15 years
Grace Period	0 year	2 year	3.5 years	0.5 year
Repayment	Equal semiannual installments	Mortgage style installments	Equal semiannual installments	Equal semiannual installments
Principal Reduction	60% at closing	50% at closing	50% at closing	50% at closing and 50% at completion of the sixth review of the ECF-supported
Past Due and Accrued Interest	60% reduction, remaining 40% capitalized	50% reduction , remaining 50% capitalized	50% reduction, remaining 50% capitalized	100% capitalized
Memo:				
NPV Haircut (12 % exit yield)	66%	68%	58%	54%
NPV Haircut (5 % exit yield)	54%	48%	39%	32%

1/ Exclude NIS holdings of the EC$2025 bond.

B. Impact of Debt Restructuring Progress on Grenada's Debt Sustainability

38. The progress achieved to date in the restructuring of Grenada's public debt marks a critical milestone in the restoration of Grenada's debt sustainability. After peaking at about 107½ percent of GDP in 2013, public debt is now expected to decline to 82½ percent of GDP by 2017 (from 99.6 percent of GDP at the first review). While the debt restructuring agreements reached to date with Grenada's creditors are anticipated to contribute about 15 percentage points out of the 22½ percent of GDP reduction in public debt over 2014-17, continued steadfast implementation of the programmed fiscal consolidation remains essential to put debt on a firmly downward path and is expected to contribute over 7 percentage points to the improvement in the debt to GDP ratio over the program period. Completion of the broader debt restructuring and continued fiscal prudence over the medium-term remain essential to restoring debt sustainability and reducing debt towards 60 percent of GDP by 2020.

2. **Grenada's risk of external debt distress has improved with progress in the restructuring of its public debt, although it will remain classified as "in distress" until all debt restructuring steps are completed and official arrears are cleared.** Combined with the programmed fiscal consolidation, the debt restructuring agreements reached thus far with Grenada's creditors are expected to reduce public external debt to 43.2 percent of GDP by 2020 (65.1 percent at the first review), from 69.9 percent of GDP in 2014.[12] In the baseline scenario of the DSA update, only the debt threshold on the present value of the debt-to-GDP ratio and the debt-service to revenue ratios are breached, with the breaches relatively minor and temporary in nature. In contrast, all debt and debt service thresholds had protracted breaches in the DSA update at the first review. These developments would warrant an improvement in Grenada's risk of external debt from "in distress" to "high risk", but the classification of "in distress" is maintained for now until all debt restructuring steps are completed and official arrears are cleared. Thus, completion of the broader restructuring of public external debt—expected before end-2015—remains critical to further reducing Grenada's risk of external debt distress.

[12] Private external debt, estimated at 78 percent of GDP in 2014, however, remains elevated and continues to present an important risk to external debt sustainability.

Table AII: 1. Grenada: Public Debt Sustainability Framework, 2012-35

(In percent of GDP)

	Actual					Estimate								Projections		
	2012	2013	2014	Average 5/	Standard Deviation 5/	2015	2016	2017	2018	2019	2020	2021	2014-20 Average	2025	2035	2021-35 Average
Public sector debt 1/	103.3	107.5	104.9			96.9	92.6	82.5	77.7	73.0	68.3	64.5		55.1	55.1	
of which: foreign-currency denominated	67.6	70.6	69.9			65.5	64.4	55.8	51.6	47.3	43.2	40.5		35.5	43.1	
Change in public sector debt	2.6	4.2	-2.6			-8.0	-4.3	-10.2	-4.8	-4.7	-4.6	-3.9		-1.4	0.0	
Identified debt-creating flows	1.0	2.9	-1.8			-8.2	-4.3	-10.2	-4.7	-4.7	-4.7	-3.9		-1.3	-0.1	
Primary deficit	2.5	4.0	1.2	2.4	1.9	-1.3	-3.5	-3.5	-3.5	-3.5	-3.5	-3.0	-3.1	-0.6	0.8	-0.1
Revenue and grants	20.8	21.0	25.5			24.2	23.9	24.7	24.8	24.9	24.9	24.9		24.9	24.9	
of which: grants	1.1	1.4	4.3			3.2	2.2	2.2	2.2	2.2	2.2	2.2		2.2	2.2	
Primary (noninterest) expenditure	23.3	25.0	26.7			22.9	20.4	21.3	21.4	21.4	21.4	21.9		24.3	25.7	
Automatic debt dynamics	0.7	-1.1	-1.0			1.1	-0.8	-0.8	-1.3	-1.3	-1.2	-0.9		-0.8	-0.9	
Contribution from interest rate/growth differential	2.2	-0.8	-0.9			1.2	0.1	-0.4	-1.0	-1.0	-0.9	-0.4		-0.5	-0.6	
of which: contribution from average real interest rate	1.0	1.7	2.2			2.9	2.0	1.5	0.9	0.9	0.9	1.3		0.9	0.7	
of which: contribution from real GDP growth	1.2	-2.4	-3.1			-1.8	-2.0	-1.8	-1.9	-1.9	-1.8	-1.7		-1.4	-1.3	
Contribution from real exchange rate depreciation	-1.4	-0.3	-0.1			-0.1	-0.9	-0.5	-0.3	-0.3	-0.3	-0.5		...	...	
Other identified debt-creating flows	-2.2	0.0	-2.0			-8.0	0.0	-5.9	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	-2.2	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other) 6/	0.0	0.0	-2.0			-6.3	0.0	-5.9	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			-1.7	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	1.7	1.3	-0.8			0.2	0.1	0.0	0.0	0.0	0.0	0.0		-0.1	0.1	
Other Sustainability Indicators																
PV of public sector debt	...	...	88.3			78.7	72.9	63.8	59.0	54.3	49.9	45.8		35.0	35.8	
of which: foreign-currency denominated	...	...	53.2			47.3	44.7	37.2	32.9	28.7	24.7	21.9		15.3	23.8	
of which: external	...	...	53.2			47.3	44.7	37.2	32.9	28.7	24.7	21.9		15.3	23.8	
PV of contingent liabilities (not included in public sector debt)7/	...	...	...			0.2	...	...	...	...	...	...		...	...	
Gross financing need 2/	5.9	7.3	23.1			21.0	20.8	18.7	18.6	18.9	18.8	19.4		17.5	12.5	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	345.6			325.3	305.1	258.0	237.9	218.4	200.5	184.3		140.6	143.9	
PV of public sector debt-to-revenue ratio (in percent)	...	...	414.7			374.1	336.0	283.2	261.0	239.5	220.0	202.1		154.3	157.9	
of which: external 3/	...	...	250.1			224.8	205.8	165.0	145.4	126.4	109.0	96.5		67.5	104.9	
Debt service-to-revenue and grants ratio (in percent) 4/	16.4	15.9	85.8			92.3	101.7	89.7	88.9	89.7	89.6	89.8		72.6	46.8	
Debt service-to-revenue ratio (in percent) 4/	17.3	17.0	102.9			106.1	112.0	98.5	97.6	98.4	98.3	98.5		79.7	51.4	
Primary deficit that stabilizes the debt-to-GDP ratio	-1.0	0.9	0.8			-1.3	0.7	0.6	1.1	1.1	1.0	0.8		0.7	0.9	
Key macroeconomic and fiscal assumptions																
Real GDP growth (in percent)	-1.1	2.4	3.0	1.4	5.5	1.7	2.1	2.0	2.4	2.5	2.5	2.5	2.2	2.5	2.5	2.5
Average nominal interest rate on forex debt (in percent)	2.7	2.6	3.0	2.0	0.6	3.1	3.3	3.2	2.8	2.8	2.7	2.7	3.0	2.7	2.7	2.7
Average real interest rate on domestic debt (in percent)	1.1	2.8	3.1	1.3	2.9	3.8	2.6	2.6	2.4	2.5	2.5	2.8	2.7	2.8	2.8	2.8
Real exchange rate depreciation (in percent, + indicates depreciation)	-2.0	-0.5	-0.1	-0.5	2.1	-0.1	...	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	3.9	2.0	1.6	2.5	2.4	0.8	2.8	2.8	2.8	2.8	2.8	2.5	2.5	2.5	2.5	2.5
Growth of real primary spending (deflated by GDP deflator, in percent)	-12.6	10.1	10.0	0.8	6.3	-12.8	-8.9	6.2	2.8	2.8	2.5	4.8	-1.2	5.3	2.5	3.8

Sources: Country authorities; and staff estimates and projections.

1/ Central government and central government guaranteed debt only.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

6/ Debt relief includes savings from the restructuring agreement reached with Eximbank in 2014 and with the creditor committee representing the majority of creditors for Grenada's 2025 bonds in 2015.

7/ Includes severance costs associated with the restructuring of state-owned enterprises.

Table AII: 2. Grenada: External Debt Sustainability Framework, 2012-35

(In percent of GDP)

	Actual				Historical 6/ Average	Standard 6/ Deviation	Projections									
	2011	2012	2013	2014			2015	2016	2017	2018	2019	2020	2014-2020 Average	2025	2035	2021-2035 Average
External debt (nominal) 1/	**136.3**	**141.0**	**145.1**	**148.0**			**143.5**	**139.9**	**130.2**	**125.5**	**119.8**	**114.2**		**99.9**	**91.2**	
of which: public and publicly guaranteed (PPG)	69.0	67.6	70.6	69.9			65.5	64.4	55.8	51.6	47.3	43.2		35.5	43.1	
Change in external debt	12.9	4.6	4.1	2.9			-4.5	-3.7	-9.7	-4.7	-5.6	-5.6		-1.8	-1.6	
Identified net debt-creating flows	16.9	13.5	3.8	6.8			6.5	2.9	2.1	1.7	1.2	1.3		2.6	2.7	
Non-interest current account deficit	**22.0**	**19.0**	**21.5**	**15.8**	**23.2**	**4.1**	**12.7**	**13.3**	**13.8**	**14.1**	**14.0**	**13.9**		**14.3**	**13.8**	14.2
Deficit in balance of goods and services	22.6	20.6	22.0	15.7			11.7	12.2	12.5	12.7	12.8	12.8		12.7	12.0	
Exports	28.2	28.9	28.7	30.4			31.8	32.1	33.4	33.7	33.9	34.1		34.6	36.1	
Imports	50.8	49.5	50.8	46.2			43.5	44.4	45.9	46.4	46.7	46.9		47.3	48.0	
Net current transfers (negative = inflow)	-3.1	-3.8	-2.2	-2.5	-4.6	2.8	-2.1	-2.1	-2.0	-2.0	-2.0	-2.1		-1.9	-1.6	-1.8
of which: official	-0.9	-1.4	-0.2	-0.6			-0.2	-0.2	-0.2	-0.2	-0.2	-0.2		-0.1	-0.1	
Other current account flows (negative = net inflow)	2.5	2.2	1.7	2.6			3.0	3.1	3.3	3.3	3.3	3.2		3.6	3.5	
Net FDI (negative = inflow)	**-5.5**	**-3.9**	**-13.5**	**-4.8**	**-10.9**	**5.4**	**-4.8**	**-5.8**	**-7.5**	**-7.6**	**-7.9**	**-7.9**		**-7.9**	**-7.8**	-7.9
Endogenous debt dynamics 2/	**0.4**	**-1.5**	**-4.2**	**-4.2**			**-1.4**	**-4.5**	**-4.2**	**-4.8**	**-4.8**	**-4.7**		**-3.8**	**-3.3**	
Contribution from nominal interest rate	1.6	2.1	1.9	2.2			2.3	2.3	2.2	1.7	1.6	1.4		1.1	1.2	
Contribution from real GDP growth	-0.9	1.5	-3.3	-4.1			-2.5	-2.8	-2.7	-3.0	-3.0	-2.8		-2.4	-2.2	
Contribution from price and exchange rate changes	-0.3	-5.1	-2.8	-2.2			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-3.9**	**-8.9**	**0.3**	**-3.9**			**-11.0**	**-6.6**	**-11.7**	**-6.4**	**-6.9**	**-7.0**		**-4.3**	**-4.3**	
of which: exceptional financing	0.0	-0.2	-0.8	-3.6			-2.9	-2.7	-0.3	0.0	0.0	0.0		0.0	0.0	
debt restructuring	0.0	0.0	0.0	-2.0			-5.4	0.0	-5.0	0.0	0.0	0.0		0.0	0.0	
capital transfers							-5.8	-4.8	-6.8	-7.0	-7.1	-7.1		-2.2	-2.2	
PV of external debt 4/	...	...	...	131.4			125.3	120.2	111.6	106.8	101.2	95.7		79.8	71.9	
In percent of exports	...	...	...	431.8			394.5	373.8	334.0	316.9	298.3	280.9		230.4	199.4	
PV of PPG external debt	**...**	**...**	**...**	**53.2**			**47.3**	**44.7**	**37.2**	**32.9**	**28.7**	**24.7**		**15.3**	**23.8**	
In percent of exports	**...**	**...**	**...**	**175.0**			**148.9**	**139.0**	**111.4**	**97.6**	**84.6**	**72.5**		**44.2**	**65.9**	
In percent of government revenues	**...**	**...**	**...**	**250.1**			**224.8**	**205.8**	**165.0**	**145.4**	**126.4**	**109.0**		**67.5**	**104.9**	
Debt service-to-exports ratio (in percent)	**5.6**	**7.2**	**6.5**	**12.6**			**14.2**	**13.9**	**12.8**	**11.9**	**11.3**	**10.0**		**7.2**	**5.9**	
PPG debt service-to-exports ratio (in percent)	**4.4**	**6.4**	**5.9**	**12.0**			**13.6**	**13.3**	**12.3**	**11.3**	**10.7**	**9.4**		**6.7**	**5.7**	
PPG debt service-to-revenue ratio (in percent)	**6.1**	**9.4**	**8.6**	**17.1**			**20.6**	**19.8**	**18.2**	**16.9**	**16.1**	**14.2**		**10.3**	**9.0**	
Total gross financing need (Millions of U.S. dollars)	140.8	137.0	82.4	129.6			110.6	112.4	104.4	109.1	108.2	108.6		131.0	196.2	
Non-interest current account deficit that stabilizes debt ratio	9.0	14.3	17.4	12.9			17.1	17.0	23.5	18.8	19.6	19.6		16.1	15.4	
Key macroeconomic assumptions																
Real GDP growth (in percent)	0.8	-1.1	2.4	3.0	1.4	5.5	1.7	2.1	2.0	2.4	2.5	2.5	2.2	2.5	2.5	2.5
GDP deflator in US dollar terms (change in percent)	0.2	3.9	2.0	1.6	2.5	2.4	0.8	2.8	2.8	2.8	2.8	2.8	2.5	2.5	2.5	2.5
Effective interest rate (percent) 5/	1.3	1.6	1.4	1.6	1.9	0.6	1.6	1.7	1.6	1.4	1.3	1.2	1.4	1.1	1.4	1.2
Growth of exports of G&S (US dollar terms, in percent)	7.3	5.2	3.8	10.8	3.6	13.8	7.0	6.2	8.9	6.2	6.1	5.9	6.5	5.4	5.8	5.5
Growth of imports of G&S (US dollar terms, in percent)	4.2	0.0	7.2	-4.9	2.9	11.1	-3.4	7.0	8.4	6.5	5.9	5.9	5.1	5.2	5.2	5.2
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	...	33.8	40.7	22.9	48.5	52.7	38.6	38.1	29.7	29.7	29.7
Government revenues (excluding grants, in percent of GDP)	20.3	19.7	19.7	21.3			21.0	21.7	22.5	22.6	22.7	22.7		22.7	22.7	22.7
Aid flows (in Millions of US dollars) 7/	26.1	8.8	11.6	37.2			39.7	40.5	31.5	30.6	31.9	33.2		32.4	53.1	
of which: Grants	26.1	8.8	11.6	37.2			28.3	20.7	21.7	22.8	24.0	25.3		32.4	53.1	
of which: Concessional loans	0.0	0.0	0.0	0.0			11.4	19.8	9.8	7.8	7.8	7.8		0.0	0.0	
Memorandum items:																
Nominal GDP (Millions of US dollars)	778.6	799.9	835.7	874.1			896.1	940.6	985.7	1037.6	1093.3	1152.0		1474.2	2414.5	
Nominal dollar GDP growth	1.0	2.7	4.5	4.6			2.5	5.0	4.8	5.3	5.4	5.4	4.8	5.1	5.1	5.1
PV of PPG external debt (in Millions of US dollars)				465.4			423.7	420.1	366.7	341.3	313.5	284.8		225.7	575.0	
(PVt-PVt-1)/GDPt-1 (in percent)							-4.8	-0.4	-5.7	-2.6	-2.7	-2.6	-2.9	0.0	2.2	0.9
Gross workers' remittances (Millions of US dollars)	29.0	29.7	30.5	31.2			32.0	33.2	34.6	36.1	40.6	45.2		55.4	83.2	
PV of PPG external debt (in percent of GDP + remittances)		...	...	51.4			45.7	43.1	35.9	31.8	27.6	23.8		14.8	23.0	
PV of PPG external debt (in percent of exports + remittances)		...	...	156.6			133.8	125.2	100.8	88.5	76.2	65.0		39.9	60.2	
Debt service of PPG external debt (in percent of exports + remittances)		...	...	10.7			12.2	12.0	11.1	10.3	9.7	8.4		6.1	5.2	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

Table AII: 3. Grenada: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2015-35

(In percent)

	Projections								
	2015	2016	2017	2018	2019	2020	2021	**2025**	2035
PV of debt-to GDP ratio									
Baseline	47	45	37	33	29	25	22	**15**	24
A. Alternative Scenarios									
A1. Key variables at their historical averages in 2015-2035 1/	47	49	46	46	46	47	49	**57**	84
A2. New public sector loans on less favorable terms in 2015-2035 2	47	45	38	34	30	26	23	**19**	34
B. Bound Tests									
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	47	48	42	37	32	28	25	**17**	27
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	47	48	48	44	39	35	32	**23**	25
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	47	46	39	35	30	26	23	**16**	25
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	47	45	39	35	31	27	24	**17**	24
B5. Combination of B1-B4 using one-half standard deviation shocks	47	47	45	40	36	31	28	**20**	27
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	47	63	53	46	40	35	31	**22**	34
PV of debt-to-exports ratio									
Baseline	149	139	111	98	85	73	64	**44**	66
A. Alternative Scenarios									
A1. Key variables at their historical averages in 2015-2035 1/	149	151	137	136	137	139	143	**164**	233
A2. New public sector loans on less favorable terms in 2015-2035 2	149	141	113	100	88	76	68	**56**	95
B. Bound Tests									
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	149	139	111	98	85	73	64	**44**	66
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	149	178	207	186	166	148	136	**96**	101
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	149	139	111	98	85	73	64	**44**	66
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	149	140	117	104	90	78	70	**48**	67
B5. Combination of B1-B4 using one-half standard deviation shocks	149	152	152	135	119	104	93	**65**	84
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	149	139	111	98	85	73	64	**44**	66
PV of debt-to-revenue ratio									
Baseline	225	206	165	145	126	109	96	**68**	105
A. Alternative Scenarios									
A1. Key variables at their historical averages in 2015-2035 1/	225	224	203	203	205	208	216	**250**	370
A2. New public sector loans on less favorable terms in 2015-2035 2	225	209	167	149	131	114	103	**85**	152
B. Bound Tests									
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	225	219	187	164	143	123	109	**76**	119
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	225	222	213	193	173	155	143	**102**	112
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	225	211	174	153	133	115	102	**71**	111
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	225	208	174	154	135	118	105	**74**	106
B5. Combination of B1-B4 using one-half standard deviation shocks	225	216	201	179	158	138	124	**88**	119
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	225	291	233	205	178	154	136	**95**	148

Table AII: 4. Grenada: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2015-35 (Continued)

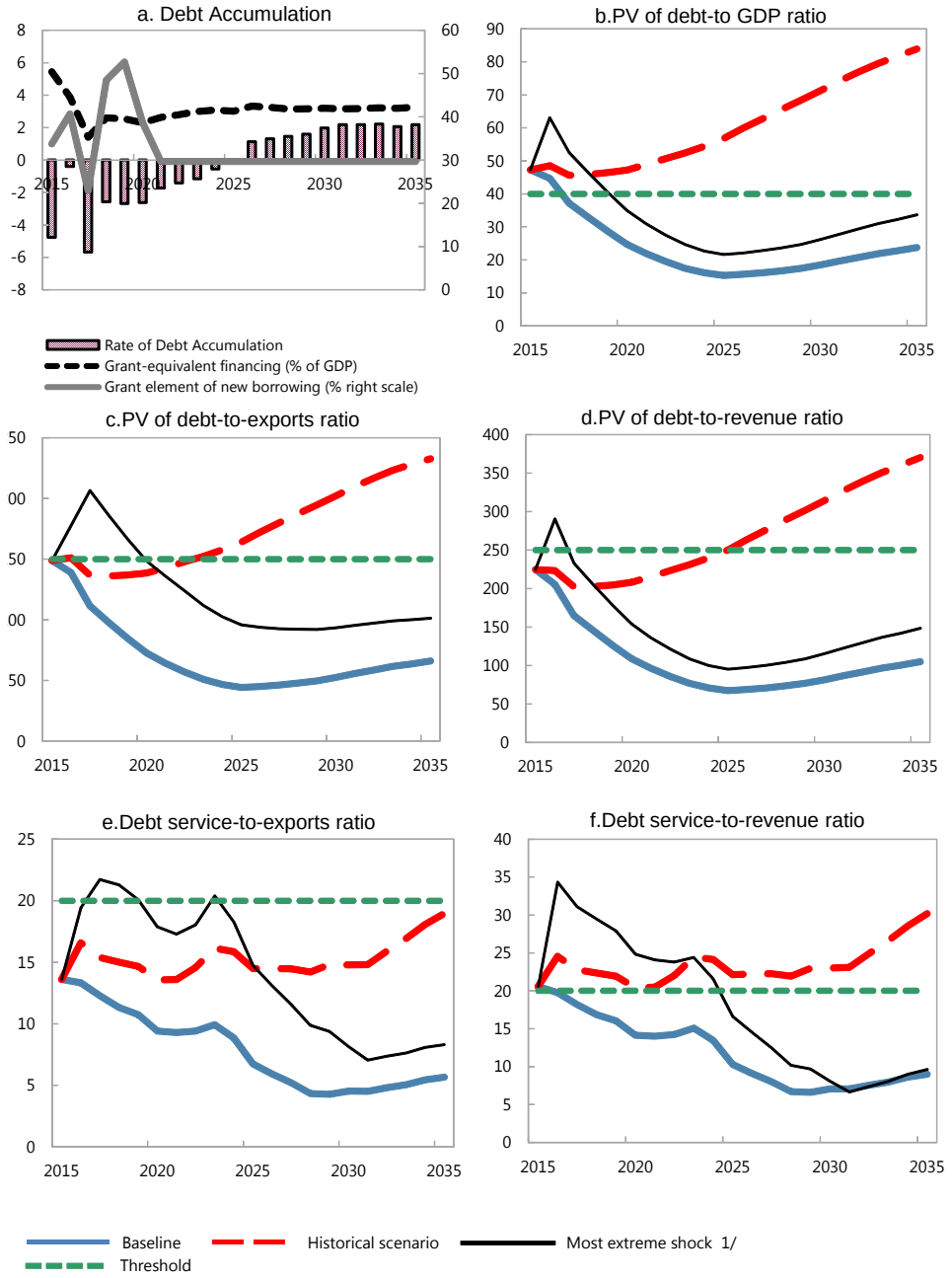
(In percent)

	Projections								
	2015	2016	2017	2018	2019	2020	2021	**2025**	2035
Debt service-to-exports ratio									
Baseline	14	13	12	11	11	9	9	**7**	6
A. Alternative Scenarios									
A1. Key variables at their historical averages in 2015-2035 1/	14	17	15	15	15	14	14	**14**	19
A2. New public sector loans on less favorable terms in 2015-2035 2	14	16	15	13	12	10	9	**6**	6
B. Bound Tests									
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	14	16	15	14	13	12	11	**8**	4
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	14	19	22	21	20	18	17	**15**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	14	16	15	14	13	12	11	**8**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	14	16	15	14	13	12	11	**8**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	14	18	18	18	17	15	14	**11**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	14	16	15	14	13	12	11	**8**	4
Debt service-to-revenue ratio									
Baseline	21	20	18	17	16	14	14	**10**	9
A. Alternative Scenarios									
A1. Key variables at their historical averages in 2015-2035 1/	21	25	23	22	22	20	20	**22**	30
A2. New public sector loans on less favorable terms in 2015-2035 2	21	24	22	20	17	16	14	**10**	10
B. Bound Tests									
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	21	26	25	24	22	20	19	**13**	8
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	21	24	22	22	21	19	18	**16**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	21	25	23	22	21	19	18	**12**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	21	24	22	21	20	18	17	**12**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	21	26	24	23	22	20	19	**15**	8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	21	34	31	29	28	25	24	**17**	10
Memorandum item:									
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	28	28	28	28	28	28	28	**28**	28

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

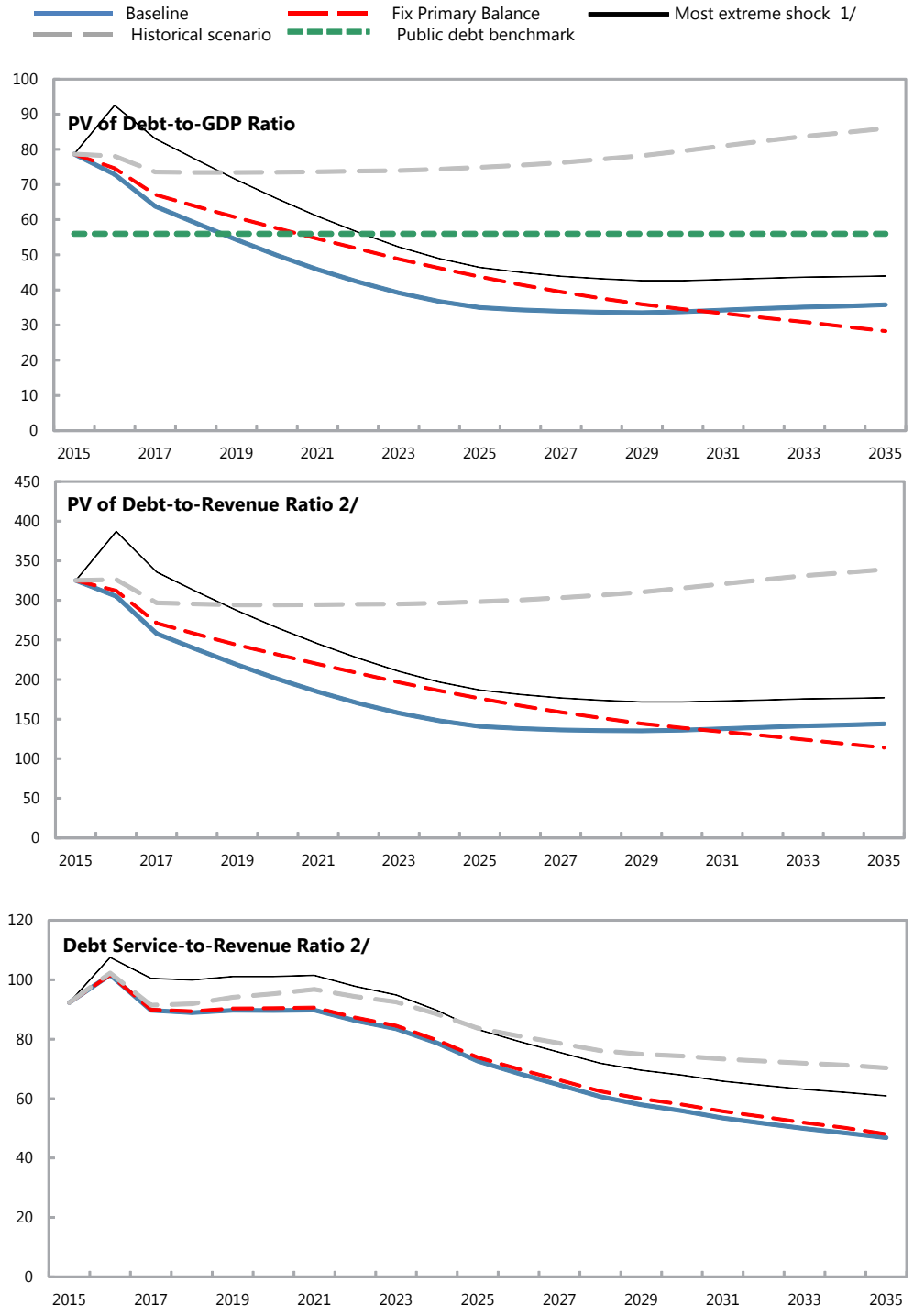
Figure AII: 1. Grenada: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2015-35 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2025. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure AII: 2. Grenada: Indicators of Public Debt under Alternative Scenarios, 2015-35 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2025.
2/ Revenues are defined inclusive of grants.